
1. Struktur folder dokumentasi

Di VM 1, buat folder dokumen final:

```
mkdir -p ~/ufosi-ops/final-documentation
mkdir -p ~/ufosi-ops/final-documentation/word-content
mkdir -p ~/ufosi-ops/final-documentation/screenshots
mkdir -p ~/ufosi-ops/final-documentation/evidence-pack
```

Folder ini isinya:

word-content = template isi Word/Docs
screenshots = tempat screenshot kalau kamu copy manual
evidence-pack = file txt/log/evidence pendukung

2. DOC-001 — Project Overview

Isi dokumen ini untuk menjelaskan UFOSI itu apa.

DOC-001: UFOSI Project Overview

1. Project Name

UFOSI - Universal Financing Optimized Syariah Intelligent-assistant

2. Project Background

UFOSI dikembangkan untuk membantu proses analisis pembiayaan, khususnya pada pengolahan dokumen rekening koran nasabah. Proses manual sebelumnya membutuhkan waktu lama, rentan human error, dan bergantung pada analisis individu. UFOSI bertujuan mempercepat proses ekstraksi, analisis, dan penyusunan hasil analisis dokumen pembiayaan secara lebih cepat, akurat, dan terdokumentasi.

3. Main Objective

Mendukung proses pembiayaan dengan sistem berbasis web, backend API, AI/OCR engine, dan database PostgreSQL yang berjalan di atas Kubernetes/K3s.

4. Main Components

- Frontend: Web UI untuk user
- Backend API: Service utama untuk authentication, data, dan integrasi aplikasi
- AI/OCR: Engine ekstraksi dan analisis rekening koran
- PostgreSQL: Database aplikasi
- Monitoring: Prometheus dan Grafana

5. Deployment Environment

- Environment: Demo / UAT
- Application Server: VM 1 / ubuntu-itops1
- IP Address: 192.168.23.72
- Platform: Docker, Kubernetes/K3s
- Namespace Application: ufosi
- Namespace Monitoring: monitoring

6. Final Version

- Frontend: gmaaliki/ufosi-fe:v1.1.1
- Backend: gmaaliki/ufosi-be:v1.1.1
- AI/OCR: gmaaliki/ufosi-ai:v1.1.0
- Database: postgres:16

7. Final Status

UFOSI latest deployment is ready for Demo/UAT from IT Ops perspective.

3. DOC-002 — Deployment Architecture

Dokumen ini untuk menjelaskan arsitektur deployment.

DOC-002: UFOSI Deployment Architecture

1. Architecture Overview

UFOSI berjalan pada VM 1 menggunakan Kubernetes/K3s. Komponen aplikasi dipisahkan menjadi frontend, backend API, AI/OCR service, database PostgreSQL, serta monitoring stack.

2. Deployment Host

- Hostname: ubuntu-itops1
- IP Address: 192.168.23.72
- Role: Application Server
- Platform: Kubernetes/K3s

3. Kubernetes Namespace

- Application Namespace: ufosi
- Monitoring Namespace: monitoring

4. Application Component

Component	Image	Service Type	Port
Frontend	gmaaliki/ufosi-fe:v1.1.1	NodePort	30080
Backend	gmaaliki/ufosi-be:v1.1.1	NodePort / ClusterIP	30081 / 8080
AI/OCR	gmaaliki/ufosi-ai:v1.1.0	ClusterIP	8000
PostgreSQL	postgres:16	ClusterIP	5432

5. Monitoring Component

Component	Access
Grafana	http://192.168.23.72:30300
Prometheus	http://192.168.23.72:30090

6. Access Flow

User accesses the frontend through NodePort 30080. Frontend communicates with backend API through NodePort 30081. Backend communicates internally with AI/OCR and PostgreSQL database inside Kubernetes namespace.

7. Security Note

Backend, AI/OCR, and PostgreSQL are not directly exposed to users except backend NodePort 30081 for UAT/demo frontend-browser access. Database remains internal.

`tkubectl get svc -n ufosi.`

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl get svc -n ufosi
NAME                TYPE                CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
db                  ClusterIP           10.43.36.123   <none>          5432/TCP          3d4h
db-test             ClusterIP           10.43.31.211   <none>          5432/TCP          3d4h
ufosi-ai            ClusterIP           10.43.19.79    <none>          8000/TCP          3d4h
ufosi-backend       ClusterIP           10.43.55.223   <none>          8080/TCP          5d1h
ufosi-backend-nodeport NodePort            10.43.132.1    <none>          8080:30081/TCP    27h
ufosi-db            ClusterIP           10.43.97.166   <none>          5432/TCP          5d1h
ufosi-frontend      NodePort            10.43.37.97    <none>          80:30080/TCP      5d1h
itops1@ubuntu-itops1:~/ufosi-ops$
```

`itops1@ubuntu-itops1:~/ufosi-ops$ kubectl get svc -n ufosi`

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
db	ClusterIP	10.43.36.123	<none>	5432/TCP	3d4h
db-test	ClusterIP	10.43.31.211	<none>	5432/TCP	3d4h
ufosi-ai	ClusterIP	10.43.19.79	<none>	8000/TCP	3d4h
ufosi-backend	ClusterIP	10.43.55.223	<none>	8080/TCP	5d1h
ufosi-backend-nodeport	NodePort	10.43.132.1	<none>	8080:30081/TCP	27h
ufosi-db	ClusterIP	10.43.97.166	<none>	5432/TCP	5d1h
ufosi-frontend	NodePort	10.43.37.97	<none>	80:30080/TCP	5d1h

itops1@ubuntu-itops1:~/ufosi-ops\$

4. DOC-003 — Kubernetes Deployment Evidence

Ini untuk bukti deployment Kubernetes.

Command evidence:

```
mkdir -p ~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes
```

```
kubectrl get pods -n ufosi -o wide \
> ~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes/pods-ufosi.txt
```

```
kubectrl get svc -n ufosi \
> ~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes/services-ufosi.txt
```

```
kubectrl get endpoints -n ufosi \
> ~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes/endpoints-ufosi.txt
```

```
kubectrl get pods -n monitoring -o wide \
> ~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes/pods-monitoring.txt
```

```
kubectrl get svc -n monitoring \
> ~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes/services-monitoring.txt
```

```
cat ~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes/pods-ufosi.txt
cat ~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes/services-ufosi.txt
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ cat ~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes/pods-ufosi.txt
cat ~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes/services-ufosi.txt
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE             NOMINATED NODE
READINESS GATES
ufosi-ai-98d44cb97-zlph2            1/1     Running   0           139m   10.42.0.201   ubuntu-itops1    <none>
<none>
ufosi-backend-7c7bc64bb4-29bp8     1/1     Running   3 (35m ago) 54m    10.42.0.208   ubuntu-itops1    <none>
<none>
ufosi-db-5c745b9db8-vk8q2          1/1     Running   7 (25h ago) 3d3h   10.42.0.172   ubuntu-itops1    <none>
<none>
ufosi-frontend-767db749dc-cbmww4    1/1     Running   0           103m   10.42.0.204   ubuntu-itops1    <none>
<none>
NAME                                TYPE               CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
db                                  ClusterIP          10.43.36.123   <none>         5432/TCP          3d4h
db-test                            ClusterIP          10.43.31.211   <none>         5432/TCP          3d4h
ufosi-ai                           ClusterIP          10.43.19.79    <none>         8080/TCP          3d3h
ufosi-backend                       ClusterIP          10.43.55.223   <none>         8080/TCP          5d1h
ufosi-backend-nodeport              NodePort           10.43.132.1    <none>         8080:30081/TCP    27h
ufosi-db                            ClusterIP          10.43.97.166   <none>         5432/TCP          5d1h
ufosi-frontend                      NodePort           10.43.37.97    <none>         80:30080/TCP      5d1h
itops1@ubuntu-itops1:~/ufosi-ops$
```

Figure. Kubernetes Deployment Status

Description: Shows UFOSI application pods and services running in namespace ufosi.

Result: PASS

```
itops1@ubuntu-itops1:~/ufosi-ops$ cat
~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes/pods-ufosi.txt
cat ~/ufosi-ops/final-documentation/evidence-pack/doc-003-kubernetes/services-ufosi.txt
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE             NOMINATED NODE
READINESS GATES
ufosi-ai-98d44cb97-zlph2            1/1     Running   0           139m   10.42.0.201   ubuntu-itops1    <none>
<none>                                <none>
```

```
ufosi-backend-7cfbc64bb4-29bp8 1/1 Running 3 (35m ago) 54m 10.42.0.208
ubuntu-itops1 <none> <none>
ufosi-db-5c745b9db8-vk8q2 1/1 Running 7 (25h ago) 3d3h 10.42.0.172
ubuntu-itops1 <none> <none>
ufosi-frontend-767db749dc-cbmw4 1/1 Running 0 103m 10.42.0.204
ubuntu-itops1 <none> <none>
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
db ClusterIP 10.43.36.123 <none> 5432/TCP 3d4h
db-test ClusterIP 10.43.31.211 <none> 5432/TCP 3d4h
ufosi-ai ClusterIP 10.43.19.79 <none> 8000/TCP 3d3h
ufosi-backend ClusterIP 10.43.55.223 <none> 8080/TCP 5d1h
ufosi-backend-nodeport NodePort 10.43.132.1 <none> 8080:30081/TCP 27h
ufosi-db ClusterIP 10.43.97.166 <none> 5432/TCP 5d1h
ufosi-frontend NodePort 10.43.37.97 <none> 80:30080/TCP 5d1h
itops1@ubuntu-itops1:~/ufosi-ops$
```

5. DOC-004 — Service Access & Endpoint List

DOC-004: UFOSI Service Access & Endpoint List

1. Frontend Access

Frontend URL:

<http://192.168.23.72:30080>

2. Backend API Access

Backend Base URL:

<http://192.168.23.72:30081>

Backend Healthcheck:

GET <http://192.168.23.72:30081/api/v1/health>

Register API:

POST <http://192.168.23.72:30081/api/v1/auth/register>

Login API:

POST <http://192.168.23.72:30081/api/v1/auth/login>

3. Authenticated GET DB Endpoints

The following endpoints require Authorization Bearer token:

- GET /api/v1/users/profile
- GET /api/v1/nasabah
- GET /api/v1/banks
- GET /api/v1/jenis-pembiayaan
- GET /api/v1/analytics/summary

4. Internal AI/OCR Endpoint

AI/OCR internal healthcheck:

<http://ufosi-ai:8000/api/v1/health>

5. Database

PostgreSQL:

db:5432

Database Name:

ufo-db

6. Monitoring Access

Grafana:

<http://192.168.23.72:30300>

Prometheus:

<http://192.168.23.72:30090>

7. Notes

Bank route uses plural endpoint:

Correct: /api/v1/banks

Incorrect: /api/v1/bank

6. DOC-005 — Operational Validation Evidence

Bagian yang masuk:

1. Final image baseline
2. Frontend healthcheck
3. Backend healthcheck
4. AI/OCR healthcheck
5. DB connection
6. GET DB validation
7. Self-healing
8. Bank cleanup
9. UI dropdown

Evidence:

~/ufosi-ops/evidence/final-word-documentation
~/ufosi-ops/evidence/day11-be-v1.1.1-upgrade
~/ufosi-ops/evidence/day11-db-cleanup-bank

7. DOC-006 — Network Connectivity Requirement

Frontend: 192.168.23.72:30080
Backend: 192.168.23.72:30081
AI/OCR internal: ufosi-ai:8000
DB internal: db:5432
Grafana: 192.168.23.72:30300
Prometheus: 192.168.23.72:30090
SSH: 22

8. DOC-007 — Firewall Rule Request

ALLOW PC/Admin → VM1:22 SSH
ALLOW Browser/User → VM1:30080 Frontend
ALLOW Browser/Frontend → VM1:30081 Backend API
ALLOW Ops → VM1:30300 Grafana
ALLOW Ops → VM1:30090 Prometheus
DENY direct DB external 5432
DENY direct AI/OCR external 8000
DB tetap internal ClusterIP

```
kubectl get svc -n ufosi
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl get svc -n ufosi
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
db	ClusterIP	10.43.36.123	<none>	5432/TCP	3d4h
db-test	ClusterIP	10.43.31.211	<none>	5432/TCP	3d4h
ufosi-ai	ClusterIP	10.43.19.79	<none>	8000/TCP	3d4h
ufosi-backend	ClusterIP	10.43.55.223	<none>	8080/TCP	5d1h
ufosi-backend-nodeport	NodePort	10.43.132.1	<none>	8080:30081/TCP	27h
ufosi-db	ClusterIP	10.43.97.166	<none>	5432/TCP	5d1h
ufosi-frontend	NodePort	10.43.37.97	<none>	80:30080/TCP	5d1h

```
itops1@ubuntu-itops1:~/ufosi-ops$
```

9. DOC-008 — Monitoring & Observability Evidence

Bukti monitoring:

Command:

```
mkdir -p ~/ufosi-ops/final-documentation/evidence-pack/doc-008-monitoring
```

```
curl -i http://192.168.23.72:30090/-/ready \  
> ~/ufosi-ops/final-documentation/evidence-pack/doc-008-monitoring/prometheus-ready.txt  
2>&1
```

```
curl -I http://192.168.23.72:30300 \  
> ~/ufosi-ops/final-documentation/evidence-pack/doc-008-monitoring/grafana-header.txt  
2>&1
```

```
kubectl get pods -n monitoring -o wide \  
> ~/ufosi-ops/final-documentation/evidence-pack/doc-008-monitoring/monitoring-pods.txt
```

```
kubectl get svc -n monitoring \  
> ~/ufosi-ops/final-documentation/evidence-pack/doc-008-monitoring/monitoring-services.txt
```

cat ~/ufosi-ops/final-documentation/evidence-pack/doc-008-monitoring/prometheus-ready.txt
cat ~/ufosi-ops/final-documentation/evidence-pack/doc-008-monitoring/grafana-header.txt

Figure. Monitoring Service Validation
Description: Shows Prometheus readiness and Grafana service access validation.
Result: PASS

```
itops1@ubuntu-itops1:~/ufosi-ops$ cat ~/ufosi-ops/final-documentation/evidence-pack/doc-008-monitoring/prometheus-ready.txt
cat ~/ufosi-ops/final-documentation/evidence-pack/doc-008-monitoring/grafana-header.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total  Spent  Left  Speed
100    28  100    28    0    0  44585      0 --:--:-- --:--:-- --:--:-- 28000
HTTP/1.1 200 OK
Date: Tue, 09 Jun 2026 11:24:39 GMT
Content-Length: 28
Content-Type: text/plain; charset=utf-8

Prometheus Server is Ready.
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total  Spent  Left  Speed
  0     0    0     0    0    0      0      0 --:--:-- --:--:-- --:--:--    0
HTTP/1.1 302 Found
Cache-Control: no-store
Content-Type: text/html; charset=utf-8
Location: /login
X-Content-Type-Options: nosniff
X-Frame-Options: deny
X-Xss-Protection: 1; mode=block
Date: Tue, 09 Jun 2026 11:24:39 GMT
```

itops1@ubuntu-itops1:~/ufosi-ops\$ cat
~/ufosi-ops/final-documentation/evidence-pack/doc-008-monitoring/prometheus-ready.txt
cat ~/ufosi-ops/final-documentation/evidence-pack/doc-008-monitoring/grafana-header.txt
 % Total % Received % Xferd Average Speed Time Time Time Current
 Dload Upload Total Spent Left Speed
100 28 100 28 0 0 44585 0 --:--:-- --:--:-- --:--:-- 28000
HTTP/1.1 200 OK
Date: Tue, 09 Jun 2026 11:24:39 GMT
Content-Length: 28
Content-Type: text/plain; charset=utf-8

Prometheus Server is Ready.
 % Total % Received % Xferd Average Speed Time Time Time Current
 Dload Upload Total Spent Left Speed
 0 0 0 0 0 0 0 0 --:--:-- --:--:-- --:--:-- 0
HTTP/1.1 302 Found
Cache-Control: no-store
Content-Type: text/html; charset=utf-8
Location: /login
X-Content-Type-Options: nosniff
X-Frame-Options: deny
X-Xss-Protection: 1; mode=block
Date: Tue, 09 Jun 2026 11:24:39 GMT

itops1@ubuntu-itops1:~/ufosi-ops\$

10. DOC-009 — Self-Healing & Recovery Evidence

Isi:

Backend pod deleted manually
Deployment recreated pod
Rollout successful
Backend healthcheck returns 200 OK

Evidence:

```
cat ~/ufosi-ops/evidence/day11-be-v1.1.1-upgrade/backend-self-healing-rollout.txt
cat ~/ufosi-ops/evidence/day11-be-v1.1.1-upgrade/backend-self-healing-health.txt
```

Figure. Backend Self-Healing Evidence

Description: Shows backend pod recovery after manual deletion and healthcheck returns HTTP 200 OK.

Result: PASS

```
itops1@ubuntu-itops1:~/ufosi-ops$ cat ~/ufosi-ops/evidence/day11-be-v1.1.1-upgrade/backend-self-healing-rollout.txt
cat ~/ufosi-ops/evidence/day11-be-v1.1.1-upgrade/backend-self-healing-health.txt
deployment "ufosi-backend" successfully rolled out
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload    Total   Spent    Left   Speed
100  97  100    97    0    0  90485      0  --:--:-- --:--:-- --:--:--  97000
HTTP/1.1 200 OK
Vary: Origin
Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data;;form-action 'self';frame-ancest
ors 'self';img-src 'self' data;;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https: 'unsa
fe-inline';upgrade-insecure-requests
Cross-Origin-Opener-Policy: same-origin
Cross-Origin-Resource-Policy: cross-origin
Origin-Agent-Cluster: ?1
Referrer-Policy: no-referrer
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Content-Type-Options: nosniff
X-DNS-Prefetch-Control: off
X-Download-Options: noopen
X-Frame-Options: SAMEORIGIN
X-Permitted-Cross-Domain-Policies: none
X-XSS-Protection: 0
Content-Type: application/json; charset=utf-8
Content-Length: 97
ETag: W/"61-HoiZuzxtQMekyuWUtPAfLIj5AyA"
Date: Tue, 09 Jun 2026 10:28:53 GMT
Connection: keep-alive
Keep-Alive: timeout=5
{"success":true,"message":"UFOSI API is running","data":{"timestamp":"2026-06-09T10:28:53.599Z"}}itops1@ubuntu-itops1:~/
```

```

itops1@ubuntu-itops1:~/ufosi-ops$ cat
~/ufosi-ops/evidence/day11-be-v1.1.1-upgrade/backend-self-healing-rollout.txt
cat ~/ufosi-ops/evidence/day11-be-v1.1.1-upgrade/backend-self-healing-health.txt
deployment "ufosi-backend" successfully rolled out
% Total    % Received % Xferd Average Speed   Time    Time     Time  Current
           Dload  Upload  Total  Spent    Left  Speed
100  97 100  97   0    0 90485    0 --:--:-- --:--:-- --:--:-- 97000
HTTP/1.1 200 OK
Vary: Origin
Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action
'self';frame-ancestors 'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr
'none';style-src 'self' https: 'unsafe-inline';upgrade-insecure-requests
Cross-Origin-Opener-Policy: same-origin
Cross-Origin-Resource-Policy: cross-origin
Origin-Agent-Cluster: ?1
Referrer-Policy: no-referrer
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Content-Type-Options: nosniff
X-DNS-Prefetch-Control: off
X-Download-Options: noopen
X-Frame-Options: SAMEORIGIN
X-Permitted-Cross-Domain-Policies: none
X-XSS-Protection: 0
Content-Type: application/json; charset=utf-8
Content-Length: 97
ETag: W/"61-HoiZuzxtQMekyu4UtPAfLlj5AyA"
Date: Tue, 09 Jun 2026 10:28:53 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"success":true,"message":"UFOSI API is
running","data":{"timestamp":"2026-06-09T10:28:53.599Z"}}itops1@ubuntu-itops1:~/ufosi-op
s$

```

11. DOC-010 — Database Cleanup & Data Validation Evidence

Isi:

Problem: dropdown bank duplicate

Root cause: seed bank dijalankan berkali-kali

Action: backup DB, cleanup duplicate, keep 1 record per bank

Validation: table bank 3 rows, duplicate 0 rows, GET /api/v1/banks 200 OK

Evidence:

```
cat ~/ufosi-ops/evidence/day11-db-cleanup-bank/bank-table-after-cleanup.txt
cat ~/ufosi-ops/evidence/day11-db-cleanup-bank/bank-duplicates-after-cleanup.txt
cat ~/ufosi-ops/evidence/day11-db-cleanup-bank/get-banks-after-cleanup.txt
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ cat ~/ufosi-ops/evidence/day11-db-cleanup-bank/bank-table-after-cleanup.txt
cat ~/ufosi-ops/evidence/day11-db-cleanup-bank/bank-duplicates-after-cleanup.txt
cat ~/ufosi-ops/evidence/day11-db-cleanup-bank/get-banks-after-cleanup.txt
id | nama | created_at | updated_at
-----+-----+-----+-----
1 | Bank Arunika | 2026-06-09 10:12:14.047 | 2026-06-09 10:12:14.047
2 | Bank Kencana | 2026-06-09 10:12:14.047 | 2026-06-09 10:12:14.047
3 | Bank Nusa | 2026-06-09 10:12:14.047 | 2026-06-09 10:12:14.047
(3 rows)

nama | count
-----+-----
(0 rows)

% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 388 100 388 0 0 136k 0 --:--:-- --:--:-- --:--:-- 189k
HTTP/1.1 200 OK
Vary: Origin
Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancest
ors 'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https: 'unsa
fe-inline';upgrade-insecure-requests
Cross-Origin-Opener-Policy: same-origin
Cross-Origin-Resource-Policy: cross-origin
Origin-Agent-Cluster: ?1
Referrer-Policy: no-referrer
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Content-Type-Options: nosniff
X-DNS-Prefetch-Control: off
X-Download-Options: noopen
X-Frame-Options: SAMEORIGIN
X-Permitted-Cross-Domain-Policies: none

(0 rows)

% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 388 100 388 0 0 136k 0 --:--:-- --:--:-- --:--:-- 189k
HTTP/1.1 200 OK
Vary: Origin
Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancest
ors 'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https: 'unsa
fe-inline';upgrade-insecure-requests
Cross-Origin-Opener-Policy: same-origin
Cross-Origin-Resource-Policy: cross-origin
Origin-Agent-Cluster: ?1
Referrer-Policy: no-referrer
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Content-Type-Options: nosniff
X-DNS-Prefetch-Control: off
X-Download-Options: noopen
X-Frame-Options: SAMEORIGIN
X-Permitted-Cross-Domain-Policies: none
X-XSS-Protection: 0
Content-Type: application/json; charset=utf-8
Content-Length: 388
ETag: W/"184-WawB6cqm/hyAwwSy7x+DPcVLH3s"
Date: Tue, 09 Jun 2026 10:55:00 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"success":true,"message":"Data bank berhasil diambil","data":[{"id":1,"nama":"Bank Arunika","createdAt":"2026-06-09T10:
12:14.047Z","updatedAt":"2026-06-09T10:12:14.047Z"}, {"id":2,"nama":"Bank Kencana","createdAt":"2026-06-09T10:12:14.047Z"
,"updatedAt":"2026-06-09T10:12:14.047Z"}, {"id":3,"nama":"Bank Nusa","createdAt":"2026-06-09T10:12:14.047Z","updatedAt":"
2026-06-09T10:12:14.047Z"}]}itops1@ubuntu-itops1:~/ufosi-ops$
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ cat
~/ufosi-ops/evidence/day11-db-cleanup-bank/bank-table-after-cleanup.txt
```

cat ~/ufosi-ops/evidence/day11-db-cleanup-bank/bank-duplicates-after-cleanup.txt
cat ~/ufosi-ops/evidence/day11-db-cleanup-bank/get-banks-after-cleanup.txt

id	nama	created_at	updated_at
----	------	------------	------------

1	Bank Arunika	2026-06-09 10:12:14.047	2026-06-09 10:12:14.047
2	Bank Kencana	2026-06-09 10:12:14.047	2026-06-09 10:12:14.047
3	Bank Nusa	2026-06-09 10:12:14.047	2026-06-09 10:12:14.047

(3 rows)

nama	count
------	-------

(0 rows)

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload Total Spent Left Speed				
100	388	100	388	0	0	136k	0 --:--:-- --:--:-- --:--:-- 189k

HTTP/1.1 200 OK

Vary: Origin
Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors 'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https: 'unsafe-inline';upgrade-insecure-requests
Cross-Origin-Opener-Policy: same-origin
Cross-Origin-Resource-Policy: cross-origin
Origin-Agent-Cluster: ?1
Referrer-Policy: no-referrer
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Content-Type-Options: nosniff
X-DNS-Prefetch-Control: off
X-Download-Options: noopen
X-Frame-Options: SAMEORIGIN
X-Permitted-Cross-Domain-Policies: none
X-XSS-Protection: 0
Content-Type: application/json; charset=utf-8
Content-Length: 388
ETag: W/"184-WawB6cqm/hyAywSy7x+DPcVLH3s"
Date: Tue, 09 Jun 2026 10:55:00 GMT
Connection: keep-alive
Keep-Alive: timeout=5

```
{
  "success": true,
  "message": "Data bank berhasil diambil",
  "data": [
    {
      "id": 1,
      "nama": "Bank Arunika",
      "createdAt": "2026-06-09T10:12:14.047Z",
      "updatedAt": "2026-06-09T10:12:14.047Z"
    },
    {
      "id": 2,
      "nama": "Bank Kencana",
      "createdAt": "2026-06-09T10:12:14.047Z",
      "updatedAt": "2026-06-09T10:12:14.047Z"
    },
    {
      "id": 3,
      "nama": "Bank Nusa",
      "createdAt": "2026-06-09T10:12:14.047Z",
      "updatedAt": "2026-06-09T10:12:14.047Z"
    }
  ]
}
```

tops1@ubuntu-itops1:~/ufosi-ops\$

12. DOC-011 — Handover Runbook

Isi:

- Cara cek pod
- Cara cek service
- Cara restart frontend/backend
- Cara cek healthcheck
- Cara cek log backend
- Cara rollback deployment
- Cara backup DB
- Cara cek monitoring

DOC-011: UFOSI Operational Handover Runbook

1. Check Application Pods

kubectl get pods -n ufosi -o wide

```
2026-06-09T10:12:14.047Z]]}itops1$ kubectl get pods -n ufosi -o wide
NAME                                READY  STATUS   RESTARTS   AGE    IP             NODE             NOMINATED NODE
READINESS GATES
ufosi-ai-98d44cb97-zlph2            1/1    Running  0          143m   10.42.0.201    ubuntu-itops1    <none>
ufosi-backend-7cfbc64bb4-29bp8     1/1    Running  3 (39m ago) 58m    10.42.0.208    ubuntu-itops1    <none>
ufosi-db-5c745b9db8-vk8q2          1/1    Running  7 (25h ago) 3d3h   10.42.0.172    ubuntu-itops1    <none>
ufosi-frontend-767db749dc-cbmw4     1/1    Running  0          107m   10.42.0.204    ubuntu-itops1    <none>
itops1@ubuntu-itops1:~/ufosi-ops$
```

itops1\$ kubectl get pods -n ufosi -o wide

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE
ufosi-ai-98d44cb97-zlph2	1/1	Running	0	143m	10.42.0.201	ubuntu-itops1	<none>
ufosi-backend-7cfbc64bb4-29bp8	1/1	Running	3 (39m ago)	58m	10.42.0.208	ubuntu-itops1	<none>
ufosi-db-5c745b9db8-vk8q2	1/1	Running	7 (25h ago)	3d3h	10.42.0.172	ubuntu-itops1	<none>
ufosi-frontend-767db749dc-cbmw4	1/1	Running	0	107m	10.42.0.204	ubuntu-itops1	<none>

itops1@ubuntu-itops1:~/ufosi-ops\$

2. Check Application Services

kubectl get svc -n ufosi

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl get svc -n ufosi
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
db	ClusterIP	10.43.36.123	<none>	5432/TCP	3d4h
db-test	ClusterIP	10.43.31.211	<none>	5432/TCP	3d4h
ufosi-ai	ClusterIP	10.43.19.79	<none>	8000/TCP	3d4h
ufosi-backend	ClusterIP	10.43.55.223	<none>	8080/TCP	5d1h
ufosi-backend-nodeport	NodePort	10.43.132.1	<none>	8080:30081/TCP	27h
ufosi-db	ClusterIP	10.43.97.166	<none>	5432/TCP	5d1h
ufosi-frontend	NodePort	10.43.37.97	<none>	80:30080/TCP	5d1h

```
itops1@ubuntu-itops1:~/ufosi-ops$
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl get svc -n ufosi
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
db	ClusterIP	10.43.36.123	<none>	5432/TCP	3d4h
db-test	ClusterIP	10.43.31.211	<none>	5432/TCP	3d4h
ufosi-ai	ClusterIP	10.43.19.79	<none>	8000/TCP	3d4h
ufosi-backend	ClusterIP	10.43.55.223	<none>	8080/TCP	5d1h
ufosi-backend-nodeport	NodePort	10.43.132.1	<none>	8080:30081/TCP	27h
ufosi-db	ClusterIP	10.43.97.166	<none>	5432/TCP	5d1h
ufosi-frontend	NodePort	10.43.37.97	<none>	80:30080/TCP	5d1h

```
itops1@ubuntu-itops1:~/ufosi-ops$
```

3. Check Monitoring Services

```
kubectl get pods -n monitoring -o wide
```

```
kubectl get svc -n monitoring
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl get pods -n monitoring -o wide
```

```
kubectl get svc -n monitoring
```

NAME	NOMINATED NODE	READINESS GATES	READY	STATUS	RESTARTS	AGE	IP	NODE
alertmanager-ufosi-monitoring-kube-prom-alertmanager-0	itops1	<none>	2/2	Running	14 (25h ago)	3d4h	10.42.0.174	ubuntu-
prometheus-ufosi-monitoring-kube-prom-prometheus-0	itops1	<none>	2/2	Running	0	3h7m	10.42.0.194	ubuntu-
ufosi-monitoring-grafana-549c5fc7ff-vc784	itops1	<none>	3/3	Running	21 (25h ago)	3d4h	10.42.0.168	ubuntu-
ufosi-monitoring-kube-prom-operator-8548cdd4bb-hcsg2	itops1	<none>	1/1	Running	7 (25h ago)	3d4h	10.42.0.175	ubuntu-
ufosi-monitoring-kube-state-metrics-7845b5dbb4-mjqc2	itops1	<none>	1/1	Running	8 (25h ago)	3d4h	10.42.0.165	ubuntu-
ufosi-monitoring-prometheus-node-exporter-5trtrt	itops1	<none>	1/1	Running	8 (25h ago)	3d4h	192.168.23.72	ubuntu-

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
alertmanager-operated	ClusterIP	None	<none>	9093/TCP, 9094/TCP, 9094/UDP	3d4h
prometheus-operated	ClusterIP	None	<none>	9090/TCP	3d4h
ufosi-monitoring-grafana	NodePort	10.43.220.59	<none>	80:30300/TCP	3d4h
ufosi-monitoring-kube-prom-alertmanager	ClusterIP	10.43.249.173	<none>	9093/TCP, 8080/TCP	3d4h
ufosi-monitoring-kube-prom-operator	ClusterIP	10.43.43.142	<none>	443/TCP	3d4h
ufosi-monitoring-kube-prom-prometheus	NodePort	10.43.51.82	<none>	9090:30090/TCP, 8080:32325/TCP	3d4h
ufosi-monitoring-kube-state-metrics	ClusterIP	10.43.78.77	<none>	8080/TCP	3d4h
ufosi-monitoring-prometheus-node-exporter	ClusterIP	10.43.161.21	<none>	9100/TCP	3d4h

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl get pods -n monitoring -o wide
```

```
kubectl get svc -n monitoring
```

NAME	READY	STATUS	RESTARTS	AGE	IP
alertmanager-ufosi-monitoring-kube-prom-alertmanager-0	2/2	Running	14 (25h ago)		
3d4h 10.42.0.174 ubuntu-itops1 <none>	<none>				
prometheus-ufosi-monitoring-kube-prom-prometheus-0	2/2	Running	0	3h7m	
10.42.0.194 ubuntu-itops1 <none>	<none>				
ufosi-monitoring-grafana-549c5fc7ff-vc784	3/3	Running	21 (25h ago)	3d4h	
10.42.0.168 ubuntu-itops1 <none>	<none>				
ufosi-monitoring-kube-prom-operator-8548cdd4bb-hcsg2	1/1	Running	7 (25h ago)		
3d4h 10.42.0.175 ubuntu-itops1 <none>	<none>				
ufosi-monitoring-kube-state-metrics-7845b5dbb4-mjqc2	1/1	Running	8 (25h ago)		
3d4h 10.42.0.165 ubuntu-itops1 <none>	<none>				
ufosi-monitoring-prometheus-node-exporter-5trrt	1/1	Running	8 (25h ago)	3d4h	
192.168.23.72 ubuntu-itops1 <none>	<none>				

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)
alertmanager-operated	ClusterIP	None	<none>	
9093/TCP,9094/TCP,9094/UDP	3d4h			
prometheus-operated	ClusterIP	None	<none>	9090/TCP
3d4h				
ufosi-monitoring-grafana	NodePort	10.43.220.59	<none>	80:30300/TCP
3d4h				
ufosi-monitoring-kube-prom-alertmanager	ClusterIP	10.43.249.173	<none>	
9093/TCP,8080/TCP	3d4h			
ufosi-monitoring-kube-prom-operator	ClusterIP	10.43.43.142	<none>	443/TCP
3d4h				
ufosi-monitoring-kube-prom-prometheus	NodePort	10.43.51.82	<none>	
9090:30090/TCP,8080:32325/TCP	3d4h			
ufosi-monitoring-kube-state-metrics	ClusterIP	10.43.78.77	<none>	8080/TCP
3d4h				
ufosi-monitoring-prometheus-node-exporter	ClusterIP	10.43.161.21	<none>	
9100/TCP	3d4h			

itops1@ubuntu-itops1:~/ufosi-ops\$

4. Check Frontend Access

```
curl -I http://192.168.23.72:30080
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ curl -I http://192.168.23.72:30080
HTTP/1.1 200 OK
Vary: rsc, next-router-state-tree, next-router-prefetch, next-router-segment-prefetch, Accept-Encoding
x-nextjs-cache: HIT
x-nextjs-prerender: 1
x-nextjs-prerender: 1
x-nextjs-stale-time: 300
X-Powered-By: Next.js
Cache-Control: s-maxage=31536000
ETag: "ga5xflthlw74v"
Content-Type: text/html; charset=utf-8
Content-Length: 9247
Date: Tue, 09 Jun 2026 11:27:50 GMT
Connection: keep-alive
Keep-Alive: timeout=5
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ curl -I http://192.168.23.72:30080
HTTP/1.1 200 OK
Vary: rsc, next-router-state-tree, next-router-prefetch, next-router-segment-prefetch,
Accept-Encoding
x-nextjs-cache: HIT
x-nextjs-prerender: 1
x-nextjs-prerender: 1
x-nextjs-stale-time: 300
X-Powered-By: Next.js
Cache-Control: s-maxage=31536000
ETag: "ga5xflthlw74v"
Content-Type: text/html; charset=utf-8
Content-Length: 9247
Date: Tue, 09 Jun 2026 11:27:50 GMT
Connection: keep-alive
Keep-Alive: timeout=5
```

```
itops1@ubuntu-itops1:~/ufosi-ops$
```

5. Check Backend Health

```
curl -i http://192.168.23.72:30081/api/v1/health
```

```

itops1@ubuntu-itops1:~/ufosi-ops$ curl -i http://192.168.23.72:30081/api/v1/health
HTTP/1.1 200 OK
Vary: Origin
Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors 'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https: 'unsafe-inline';upgrade-insecure-requests
Cross-Origin-Opener-Policy: same-origin
Cross-Origin-Resource-Policy: cross-origin
Origin-Agent-Cluster: ?1
Referrer-Policy: no-referrer
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Content-Type-Options: nosniff
X-DNS-Prefetch-Control: off
X-Download-Options: noopen
X-Frame-Options: SAMEORIGIN
X-Permitted-Cross-Domain-Policies: none
X-XSS-Protection: 0
Content-Type: application/json; charset=utf-8
Content-Length: 97
ETag: W/"61-6xeObU+N7a5CmOnCLuXmMHMVrBo"
Date: Tue, 09 Jun 2026 11:28:17 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"success":true,"message":"UFOSI API is running","data":{"timestamp":"2026-06-09T11:28:17.941Z"}}itops1@ubuntu-itops1:~/ufosi-ops$

```

```

itops1@ubuntu-itops1:~/ufosi-ops$ curl -i http://192.168.23.72:30081/api/v1/health
HTTP/1.1 200 OK
Vary: Origin
Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors 'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https: 'unsafe-inline';upgrade-insecure-requests
Cross-Origin-Opener-Policy: same-origin
Cross-Origin-Resource-Policy: cross-origin
Origin-Agent-Cluster: ?1
Referrer-Policy: no-referrer
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Content-Type-Options: nosniff
X-DNS-Prefetch-Control: off
X-Download-Options: noopen
X-Frame-Options: SAMEORIGIN
X-Permitted-Cross-Domain-Policies: none
X-XSS-Protection: 0
Content-Type: application/json; charset=utf-8
Content-Length: 97
ETag: W/"61-6xeObU+N7a5CmOnCLuXmMHMVrBo"
Date: Tue, 09 Jun 2026 11:28:17 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"success":true,"message":"UFOSI API is running","data":{"timestamp":"2026-06-09T11:28:17.941Z"}}itops1@ubuntu-itops1:~/ufosi-ops$

```

6. Check AI/OCR Internal Health

```

kubectl run curl-test -n ufosi --image=curlimages/curl:latest --restart=Never -- sleep 3600
kubectl exec -n ufosi curl-test -- curl -i http://ufosi-ai:8000/api/v1/health

```

kubectl delete pod curl-test -n ufosi --ignore-not-found

```
3600-ops$ kubectl run curl-test -n ufosi --image=curlimages/curl:latest --restart=Never -- sleep 3600
kubectl exec -n ufosi curl-test -- curl -i http://ufosi-ai:8000/api/v1/health
kubectl delete pod curl-test -n ufosi --ignore-not-found
pod/curl-test created
error: Internal error occurred: unable to upgrade connection: container not found ("curl-test")
pod "curl-test" deleted from ufosi namespace
```

```
-ops$ kubectl run curl-test -n ufosi --image=curlimages/curl:latest --restart=Never -- sleep
3600
kubectl exec -n ufosi curl-test -- curl -i http://ufosi-ai:8000/api/v1/health
kubectl delete pod curl-test -n ufosi --ignore-not-found
pod/curl-test created
error: Internal error occurred: unable to upgrade connection: container not found ("curl-test")
pod "curl-test" deleted from ufosi namespace
itops1@ubuntu-itops1:~/ufosi-ops$
```

7. Check Database Connection

kubectl exec deployment/ufosi-db -n ufosi -- psql -U ufo_admin -d "ufo-db" -c '\conninfo'

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl exec deployment/ufosi-db -n ufosi -- psql -U ufo_admin -d "ufo-db" -c '\conninfo'
fo'
You are connected to database "ufo-db" as user "ufo_admin" via socket in "/var/run/postgresql" at port "5432".
itops1@ubuntu-itops1:~/ufosi-ops$
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl exec deployment/ufosi-db -n ufosi -- psql -U
ufo_admin -d "ufo-db" -c '\conninfo'
```

```
You are connected to database "ufo-db" as user "ufo_admin" via socket in
"/var/run/postgresql" at port "5432".
itops1@ubuntu-itops1:~/ufosi-ops$
```

8. Check Backend Logs

kubectl logs deployment/ufosi-backend -n ufosi --tail=200

```

itops1@ubuntu-itops1:~/ufosi-ops$ kubectl logs deployment/ufosi-backend -n ufosi --tail=200
Prisma schema loaded from prisma/schema.prisma
Datasource "db": PostgreSQL database "ufo-db", schema "public" at "db:5432"

7 migrations found in prisma/migrations

No pending migrations to apply.

> ufosi-be@1.0.0 db:seed
> ts-node prisma/seed.ts

✓ Konfigurasi bank selesai
✓ Dummy OCR – dokumen ID: b1c2d3e4-0002-0002-0002-fa1234567891 | 32 transaksi | nasabah: Ahmad Fauzan Rahardjo
✓ Seed selesai – akun admin: admin@bsi.co.id / Admin@1234

> ufosi-be@1.0.0 start
> node dist/server.js

Database connected
Server running on port 8080 in production mode
Unhandled error: Error: AI service error 500: {"error":"Internal server error during anomaly detection."}

    at AnalyticsService.detectAnomalies (/app/dist/modules/analytics/analytics.service.js:241:19)
    at process.processTicksAndRejections (node:internal/process/task_queues:104:5)
    at async detectAnomalies (/app/dist/modules/analytics/analytics.controller.js:69:26)
Unhandled error: Error: AI service error 500: {"error":"Internal server error during financial analysis."}

    at AnalyticsService.analyzeDocument (/app/dist/modules/analytics/analytics.service.js:152:19)
    at process.processTicksAndRejections (node:internal/process/task_queues:104:5)
    at async analyzeDocument (/app/dist/modules/analytics/analytics.controller.js:59:26)
Unhandled error: Error: AI service error 500: {"error":"Internal server error during anomaly detection."}

    at AnalyticsService.detectAnomalies (/app/dist/modules/analytics/analytics.service.js:241:19)

✓ Konfigurasi bank selesai
✓ Dummy OCR – dokumen ID: b1c2d3e4-0002-0002-0002-fa1234567891 | 32 transaksi | nasabah: Ahmad Fauzan Rahardjo
✓ Seed selesai – akun admin: admin@bsi.co.id / Admin@1234

> ufosi-be@1.0.0 start
> node dist/server.js

Database connected
Server running on port 8080 in production mode
Unhandled error: Error: AI service error 500: {"error":"Internal server error during anomaly detection."}

    at AnalyticsService.detectAnomalies (/app/dist/modules/analytics/analytics.service.js:241:19)
    at process.processTicksAndRejections (node:internal/process/task_queues:104:5)
    at async detectAnomalies (/app/dist/modules/analytics/analytics.controller.js:69:26)
Unhandled error: Error: AI service error 500: {"error":"Internal server error during financial analysis."}

    at AnalyticsService.analyzeDocument (/app/dist/modules/analytics/analytics.service.js:152:19)
    at process.processTicksAndRejections (node:internal/process/task_queues:104:5)
    at async analyzeDocument (/app/dist/modules/analytics/analytics.controller.js:59:26)
Unhandled error: Error: AI service error 500: {"error":"Internal server error during anomaly detection."}

    at AnalyticsService.detectAnomalies (/app/dist/modules/analytics/analytics.service.js:241:19)
    at process.processTicksAndRejections (node:internal/process/task_queues:104:5)
    at async detectAnomalies (/app/dist/modules/analytics/analytics.controller.js:69:26)
Unhandled error: Error: AI service error 500: {"error":"Internal server error during anomaly detection."}

    at AnalyticsService.detectAnomalies (/app/dist/modules/analytics/analytics.service.js:241:19)
    at process.processTicksAndRejections (node:internal/process/task_queues:104:5)
    at async detectAnomalies (/app/dist/modules/analytics/analytics.controller.js:69:26)
Unhandled error: Error: AI service error 500: {"error":"Internal server error during anomaly detection."}

    at AnalyticsService.detectAnomalies (/app/dist/modules/analytics/analytics.service.js:241:19)
    at process.processTicksAndRejections (node:internal/process/task_queues:104:5)
    at async detectAnomalies (/app/dist/modules/analytics/analytics.controller.js:69:26)
itops1@ubuntu-itops1:~/ufosi-ops$

```

```

itops1@ubuntu-itops1:~/ufosi-ops$ kubectl logs deployment/ufosi-backend -n ufosi --tail=200
Prisma schema loaded from prisma/schema.prisma
Datasource "db": PostgreSQL database "ufo-db", schema "public" at "db:5432"

```

7 migrations found in prisma/migrations

No pending migrations to apply.

```
> ufosi-be@1.0.0 db:seed
```

```
> ts-node prisma/seed.ts
```

✓ Konfigurasi bank selesai

✓ Dummy OCR — dokumen ID: b1c2d3e4-0002-0002-0002-fa1234567891 | 32 transaksi |
nasabah: Ahmad Fauzan Rahardjo

✓ Seed selesai — akun admin: admin@bsi.co.id / Admin@1234

```
> ufosi-be@1.0.0 start
```

```
> node dist/server.js
```

Database connected

Server running on port 8080 in production mode

Unhandled error: Error: AI service error 500: {"error":"Internal server error during anomaly
detection."}

at AnalyticsService.detectAnomalies

(/app/dist/modules/analytics/analytics.service.js:241:19)

at process.processTicksAndRejections (node:internal/process/task_queues:104:5)

at async detectAnomalies (/app/dist/modules/analytics/analytics.controller.js:69:26)

Unhandled error: Error: AI service error 500: {"error":"Internal server error during financial
analysis."}

at AnalyticsService.analyzeDocument

(/app/dist/modules/analytics/analytics.service.js:152:19)

at process.processTicksAndRejections (node:internal/process/task_queues:104:5)

at async analyzeDocument (/app/dist/modules/analytics/analytics.controller.js:59:26)

Unhandled error: Error: AI service error 500: {"error":"Internal server error during anomaly
detection."}

at AnalyticsService.detectAnomalies

(/app/dist/modules/analytics/analytics.service.js:241:19)

at process.processTicksAndRejections (node:internal/process/task_queues:104:5)

at async detectAnomalies (/app/dist/modules/analytics/analytics.controller.js:69:26)

Unhandled error: Error: AI service error 500: {"error":"Internal server error during anomaly
detection."}

at AnalyticsService.detectAnomalies

(/app/dist/modules/analytics/analytics.service.js:241:19)

at process.processTicksAndRejections (node:internal/process/task_queues:104:5)

at async detectAnomalies (/app/dist/modules/analytics/analytics.controller.js:69:26)

Unhandled error: Error: AI service error 500: {"error":"Internal server error during anomaly
detection."}

```
    at AnalyticsService.detectAnomalies
(/app/dist/modules/analytics/analytics.service.js:241:19)
    at process.processTicksAndRejections (node:internal/process/task_queues:104:5)
    at async detectAnomalies (/app/dist/modules/analytics/analytics.controller.js:69:26)
itops1@ubuntu-itops1:~/ufosi-ops$
```

9. Restart Frontend

```
kubectl rollout restart deployment/ufosi-frontend -n ufosi
kubectl rollout status deployment/ufosi-frontend -n ufosi --timeout=180s
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl rollout restart deployment/ufosi-frontend -n ufosi
kubectl rollout status deployment/ufosi-frontend -n ufosi --timeout=180s
deployment.apps/ufosi-frontend restarted
Waiting for deployment "ufosi-frontend" rollout to finish: 0 out of 1 new replicas have been updated...
Waiting for deployment "ufosi-frontend" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "ufosi-frontend" rollout to finish: 1 old replicas are pending termination...
deployment "ufosi-frontend" successfully rolled out
itops1@ubuntu-itops1:~/ufosi-ops$
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl rollout restart deployment/ufosi-frontend -n ufosi
kubectl rollout status deployment/ufosi-frontend -n ufosi --timeout=180s
deployment.apps/ufosi-frontend restarted
Waiting for deployment "ufosi-frontend" rollout to finish: 0 out of 1 new replicas have been
updated...
Waiting for deployment "ufosi-frontend" rollout to finish: 1 old replicas are pending
termination...
Waiting for deployment "ufosi-frontend" rollout to finish: 1 old replicas are pending
termination...
deployment "ufosi-frontend" successfully rolled out
itops1@ubuntu-itops1:~/ufosi-ops$
```

10. Restart Backend

```
kubectl rollout restart deployment/ufosi-backend -n ufosi
kubectl rollout status deployment/ufosi-backend -n ufosi --timeout=240s
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl rollout restart deployment/ufosi-backend -n ufosi
kubectl rollout status deployment/ufosi-backend -n ufosi --timeout=240s
deployment.apps/ufosi-backend restarted
Waiting for deployment "ufosi-backend" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "ufosi-backend" rollout to finish: 1 old replicas are pending termination...
deployment "ufosi-backend" successfully rolled out
itops1@ubuntu-itops1:~/ufosi-ops$
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl rollout restart deployment/ufosi-backend -n ufosi
kubectl rollout status deployment/ufosi-backend -n ufosi --timeout=240s
deployment.apps/ufosi-backend restarted
Waiting for deployment "ufosi-backend" rollout to finish: 1 old replicas are pending
termination...
Waiting for deployment "ufosi-backend" rollout to finish: 1 old replicas are pending
termination...
deployment "ufosi-backend" successfully rolled out
```

itops1@ubuntu-itops1:~/ufosi-ops\$

11. Rollback Frontend

kubectl rollout undo deployment/ufosi-frontend -n ufosi

kubectl rollout status deployment/ufosi-frontend -n ufosi --timeout=180s

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl rollout undo deployment/ufosi-frontend -n ufosi
kubectl rollout status deployment/ufosi-frontend -n ufosi --timeout=180s
deployment.apps/ufosi-frontend rolled back
Waiting for deployment "ufosi-frontend" rollout to finish: 0 out of 1 new replicas have been updated...
Waiting for deployment "ufosi-frontend" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "ufosi-frontend" rollout to finish: 1 old replicas are pending termination...
deployment "ufosi-frontend" successfully rolled out
```

itops1@ubuntu-itops1:~/ufosi-ops\$ kubectl rollout undo deployment/ufosi-frontend -n ufosi

kubectl rollout status deployment/ufosi-frontend -n ufosi --timeout=180s

deployment.apps/ufosi-frontend rolled back

Waiting for deployment "ufosi-frontend" rollout to finish: 0 out of 1 new replicas have been updated...

Waiting for deployment "ufosi-frontend" rollout to finish: 1 old replicas are pending termination...

Waiting for deployment "ufosi-frontend" rollout to finish: 1 old replicas are pending termination...

deployment "ufosi-frontend" successfully rolled out

itops1@ubuntu-itops1:~/ufosi-ops\$

12. Rollback Backend

kubectl rollout undo deployment/ufosi-backend -n ufosi

kubectl rollout status deployment/ufosi-backend -n ufosi --timeout=240s

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl rollout undo deployment/ufosi-backend -n ufosi
kubectl rollout status deployment/ufosi-backend -n ufosi --timeout=240s
deployment.apps/ufosi-backend rolled back
Waiting for deployment "ufosi-backend" rollout to finish: 0 out of 1 new replicas have been updated...
Waiting for deployment "ufosi-backend" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "ufosi-backend" rollout to finish: 1 old replicas are pending termination...
deployment "ufosi-backend" successfully rolled out
```

itops1@ubuntu-itops1:~/ufosi-ops\$ kubectl rollout undo deployment/ufosi-backend -n ufosi

kubectl rollout status deployment/ufosi-backend -n ufosi --timeout=240s

deployment.apps/ufosi-backend rolled back

Waiting for deployment "ufosi-backend" rollout to finish: 0 out of 1 new replicas have been updated...

Waiting for deployment "ufosi-backend" rollout to finish: 1 old replicas are pending termination...

Waiting for deployment "ufosi-backend" rollout to finish: 1 old replicas are pending termination...

deployment "ufosi-backend" successfully rolled out

itops1@ubuntu-itops1:~/ufosi-ops\$

13. Backup Database

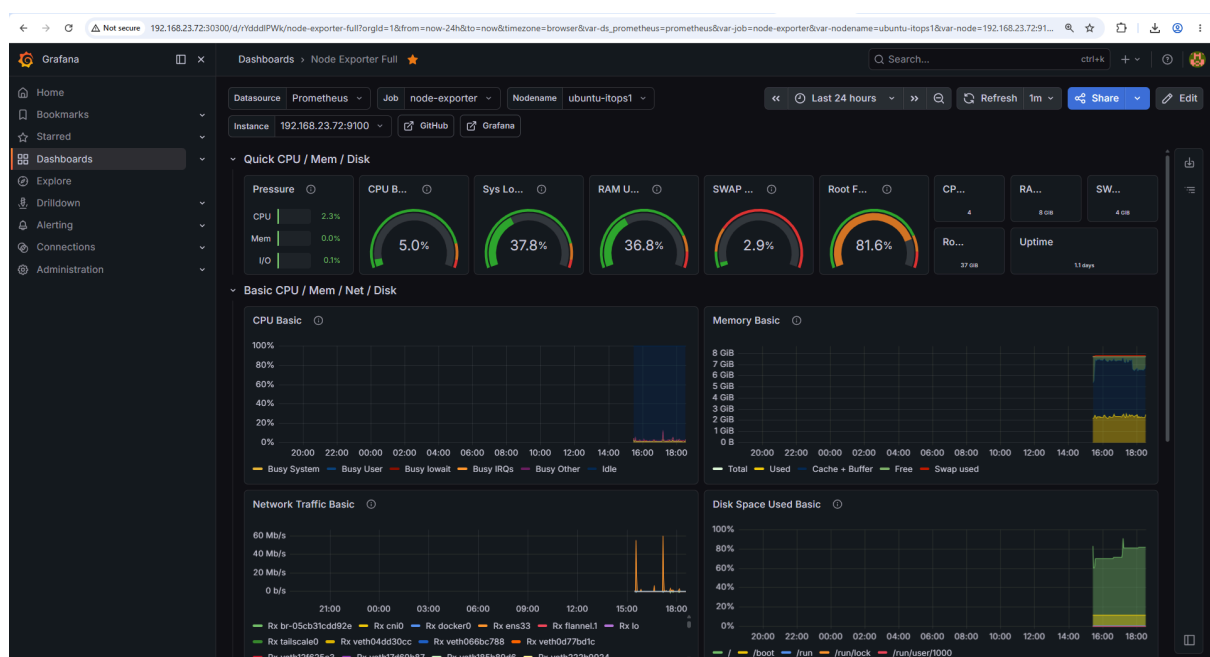
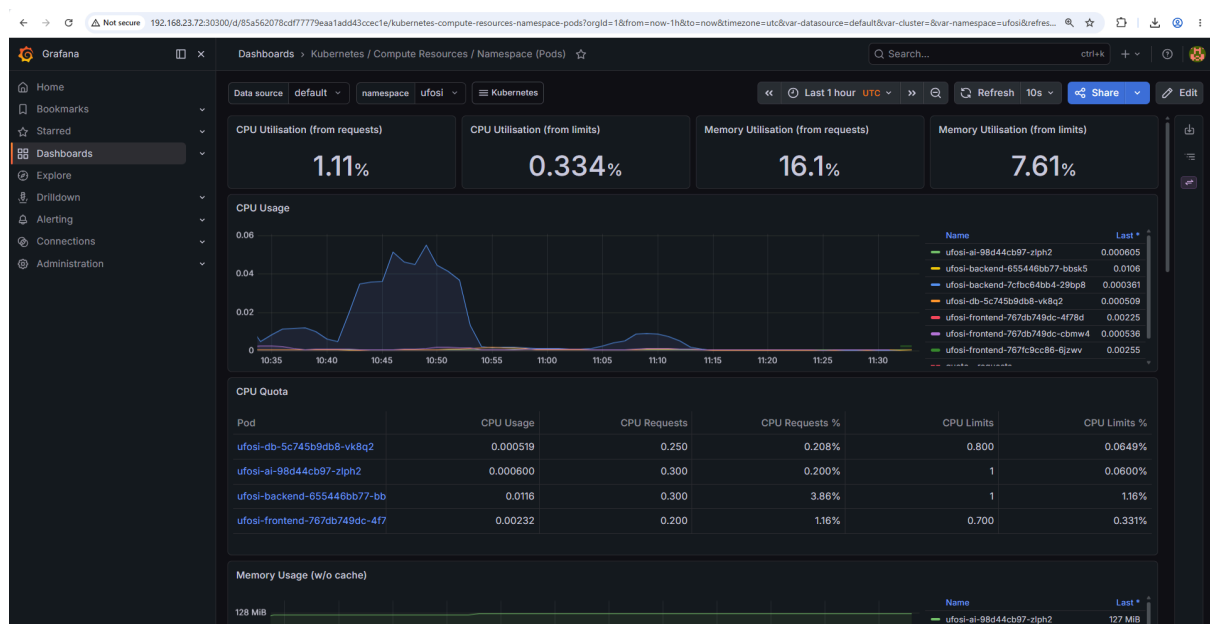
kubectl exec deployment/ufosi-db -n ufosi -- pg_dump -U ufo_admin -d "ufo-db" > ufo-db-backup.sql

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl exec deployment/ufosi-db -n ufosi -- pg_dump -U ufo_admin -d "ufo-db" > ufo-db-backup.sql
itops1@ubuntu-itops1:~/ufosi-ops$
```

14. Monitoring Access

Grafana:

<http://192.168.23.72:30300>



Prometheus:

<http://192.168.23.72:30090>

The screenshot shows the Prometheus web interface at <http://192.168.23.72:30090>. The page is titled 'Status > Target health'. It displays a list of service monitors and their targets. The first target is 'serviceMonitor/monitoring/ufosi-monitoring-grafana/0' with endpoint 'http://10.42.0.168:3000/metrics'. The second target is 'serviceMonitor/monitoring/ufosi-monitoring-kube-prom-alertmanager/0' with endpoint 'http://10.42.0.174:9093/metrics'. The third target is 'serviceMonitor/monitoring/ufosi-monitoring-kube-prom-alertmanager/1' with endpoint 'http://10.42.0.174:8080/metrics'. The fourth target is 'serviceMonitor/monitoring/ufosi-monitoring-kube-prom-apiserver/0'. Each target shows its last scrape time and state (UP).

13. DOC-012 — Final Demo/UAT Readiness Summary

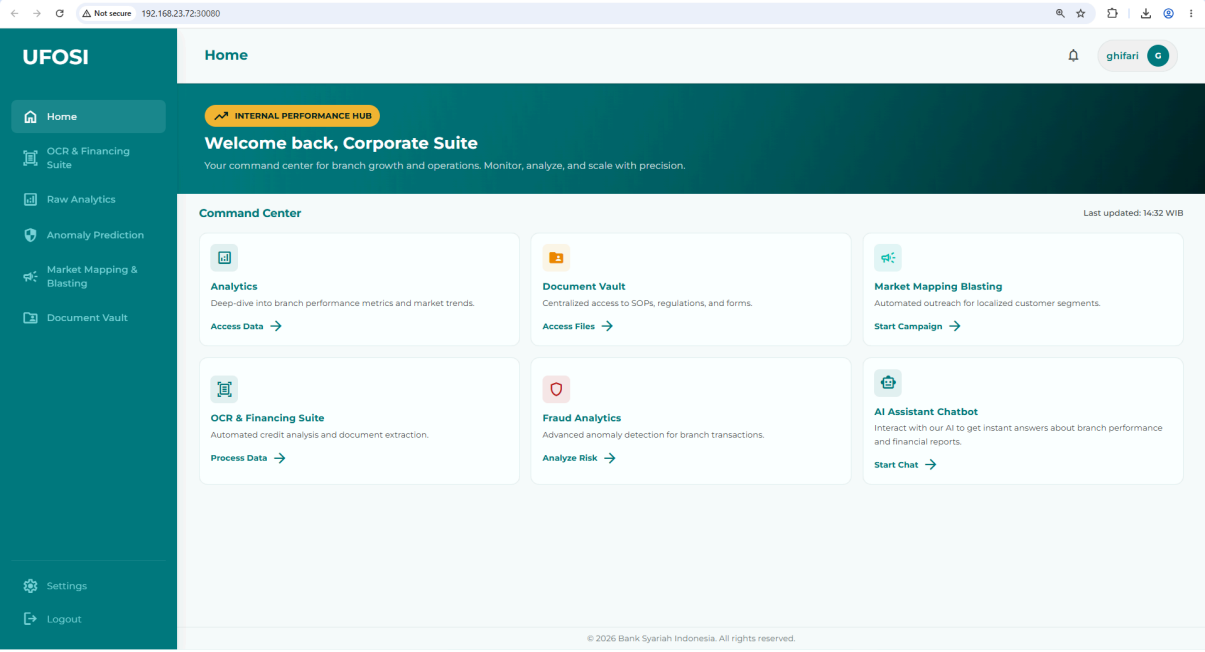
DOC-012: UFOSI Final Demo/UAT Readiness Summary

1. Final Version

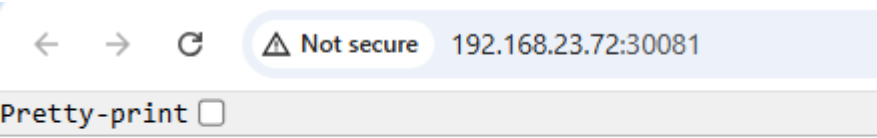
- Frontend: gmaaliki/ufosi-fe:v1.1.1
- Backend: gmaaliki/ufosi-be:v1.1.1
- AI/OCR: gmaaliki/ufosi-ai:v1.1.0
- Database: postgres:16

2. Access URL

- Frontend: <http://192.168.23.72:30080>



- Backend API: <http://192.168.23.72:30081>



- Grafana: <http://192.168.23.72:30300>
- Prometheus: <http://192.168.23.72:30090>

3. Validation Summary

Area	Result
Kubernetes pods	PASS
Frontend access	PASS
Backend healthcheck	PASS
AI/OCR internal healthcheck	PASS
PostgreSQL database connection	PASS
Register API	PASS
Login/auth token	PASS

Authenticated GET DB	PASS
Monitoring service	PASS
Backend self-healing	PASS
Bank dropdown cleanup	PASS

4. Known Notes

- Bank endpoint uses `/api/v1/banks`, not `/api/v1/bank`.
- Backend API is exposed via NodePort 30081 for Demo/UAT.
- Database remains internal inside Kubernetes.
- Secrets must not be included in public documentation.
- The shared credentials and API keys should be rotated after demo.

5. Final Decision

UFOSI latest deployment is GO for Demo/UAT from IT Ops perspective.

6. Conclusion

UFOSI has been validated successfully across deployment, service access, backend API, AI/OCR, database connectivity, monitoring, self-healing, and data cleanup areas.

Lengkapi Evidence UFW Firewall

UFW sudah aktif dan hasilnya PASS

di VM 1:

```
sudo ufw status verbose \  
> ~/ufosi-ops/evidence/day11-firewall-check/ufw-status-after-enable.txt
```

```
sudo ufw status numbered \  
> ~/ufosi-ops/evidence/day11-firewall-check/ufw-status-numbered-after-enable.txt
```

```
curl -I http://192.168.23.72:30080 \  
> ~/ufosi-ops/evidence/day11-firewall-check/frontend-after-ufw.txt 2>&1
```

```
curl -i http://192.168.23.72:30081/api/v1/health \  
> ~/ufosi-ops/evidence/day11-firewall-check/backend-after-ufw.txt 2>&1
```

```
curl -i http://192.168.23.72:30090/-/ready \  
> ~/ufosi-ops/evidence/day11-firewall-check/prometheus-after-ufw.txt 2>&1
```

```
curl -I http://192.168.23.72:30300 \  
> ~/ufosi-ops/evidence/day11-firewall-check/grafana-after-ufw.txt 2>&1
```

Cek hasil ringkas:

```
cat ~/ufosi-ops/evidence/day11-firewall-check/ufw-status-after-enable.txt  
grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/frontend-after-ufw.txt  
grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/backend-after-ufw.txt  
grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/prometheus-after-ufw.txt  
grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/grafana-after-ufw.txt
```

Status: active
Frontend: HTTP/1.1 200 OK
Backend: HTTP/1.1 200 OK
Prometheus: HTTP/1.1 200 OK
Grafana: HTTP/1.1 302 Found

Firewall / UFW Evidence: PASS Demo/UAT

```
itops1@ubuntu-itops1:~/ufosi-ops$ cat ~/ufosi-ops/evidence/day11-firewall-check/ufw-status-after-enable.txt  
grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/frontend-after-ufw.txt  
grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/backend-after-ufw.txt  
grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/prometheus-after-ufw.txt  
grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/grafana-after-ufw.txt  
Status: active  
Logging: on (low)  
Default: deny (incoming), allow (outgoing), deny (routed)  
New profiles: skip  
  
To Action From  
--  
22/tcp ALLOW IN Anywhere # Allow SSH admin access  
22/tcp on tailscale0 ALLOW IN Anywhere # Allow SSH via Tailscale  
30080/tcp ALLOW IN Anywhere # Allow UFOSI frontend NodePort  
30081/tcp ALLOW IN Anywhere # Allow UFOSI backend API NodePort  
30300/tcp ALLOW IN Anywhere # Allow Grafana monitoring  
30090/tcp ALLOW IN Anywhere # Allow Prometheus monitoring  
22/tcp (v6) ALLOW IN Anywhere (v6) # Allow SSH admin access  
22/tcp (v6) on tailscale0 ALLOW IN Anywhere (v6) # Allow SSH via Tailscale  
30080/tcp (v6) ALLOW IN Anywhere (v6) # Allow UFOSI frontend NodePort  
30081/tcp (v6) ALLOW IN Anywhere (v6) # Allow UFOSI backend API NodePort  
30300/tcp (v6) ALLOW IN Anywhere (v6) # Allow Grafana monitoring  
30090/tcp (v6) ALLOW IN Anywhere (v6) # Allow Prometheus monitoring  
  
HTTP/1.1 200 OK  
HTTP/1.1 200 OK  
HTTP/1.1 200 OK  
HTTP/1.1 302 Found
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ cat  
~/ufosi-ops/evidence/day11-firewall-check/ufw-status-after-enable.txt
```

```

grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/frontend-after-ufw.txt
grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/backend-after-ufw.txt
grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/prometheus-after-ufw.txt
grep "HTTP/1.1" ~/ufosi-ops/evidence/day11-firewall-check/grafana-after-ufw.txt
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To                Action    From
--                -
22/tcp            ALLOW IN  Anywhere          # Allow SSH admin access
22/tcp on tailscale0 ALLOW IN  Anywhere          # Allow SSH via Tailscale
30080/tcp         ALLOW IN  Anywhere          # Allow UFOSI frontend NodePort
30081/tcp         ALLOW IN  Anywhere          # Allow UFOSI backend API
NodePort
30300/tcp         ALLOW IN  Anywhere          # Allow Grafana monitoring
30090/tcp         ALLOW IN  Anywhere          # Allow Prometheus monitoring
22/tcp (v6)       ALLOW IN  Anywhere (v6)     # Allow SSH admin access
22/tcp (v6) on tailscale0 ALLOW IN  Anywhere (v6)     # Allow SSH via Tailscale
30080/tcp (v6)    ALLOW IN  Anywhere (v6)     # Allow UFOSI frontend
NodePort
30081/tcp (v6)    ALLOW IN  Anywhere (v6)     # Allow UFOSI backend API
NodePort
30300/tcp (v6)    ALLOW IN  Anywhere (v6)     # Allow Grafana monitoring
30090/tcp (v6)    ALLOW IN  Anywhere (v6)     # Allow Prometheus monitoring

HTTP/1.1 200 OK
HTTP/1.1 200 OK
HTTP/1.1 200 OK
HTTP/1.1 302 Found
itops1@ubuntu-itops1:~/ufosi-ops$

```

Fix Evidence AI/OCR Healthcheck

Ada error `container not found ("curl-test")`. Itu karena pod `curl-test` belum ready tapi sudah langsung dieksekusi.

di VM 1:

```
kubectl delete pod curl-test -n ufosi --ignore-not-found
```

```
kubectl run curl-test -n ufosi \
```

```
--image=curlimages/curl:latest \
--restart=Never \
-- sleep 3600
```

```
kubectrl wait --for=condition=Ready pod/curl-test -n ufosi --timeout=90s
```

Kalau sudah ready, healthcheck AI:

```
kubectrl exec -n ufosi curl-test -- curl -i http://ufosi-ai:8000/api/v1/health \
>
~/ufosi-ops/final-documentation/evidence-pack/doc-016-ai-healthcheck-fixed/ai-healthcheck-
fixed.txt
```

```
kubectrl delete pod curl-test -n ufosi --ignore-not-found
```

Cek hasil:

```
cat
~/ufosi-ops/final-documentation/evidence-pack/doc-016-ai-healthcheck-fixed/ai-healthcheck-
fixed.txt
```

HTTP/1.1 200 OK

AI/OCR Internal Healthcheck Evidence: PASS

```
itops1@ubuntu-itops1:~/ufosi-ops$ cat ~/ufosi-ops/final-documentation/evidence-pack/doc-016-ai-healthcheck-fixed/ai-heal
thcheck-fixed.txt
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.11.15
Date: Wed, 10 Jun 2026 01:41:45 GMT
Content-Type: application/json
Content-Length: 63
Connection: close
{"status":"ok","timestamp":"2026-06-10T01:41:45.955984+00:00"}
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ cat
~/ufosi-ops/final-documentation/evidence-pack/doc-016-ai-healthcheck-fixed/ai-healthcheck-
fixed.txt
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.11.15
Date: Wed, 10 Jun 2026 01:41:45 GMT
Content-Type: application/json
Content-Length: 63
Connection: close
```

```
{"status":"ok","timestamp":"2026-06-10T01:41:45.955984+00:00"}
itops1@ubuntu-itops1:~/ufosi-ops$
```

Update Gemini API Key Baru

IT Developer sudah kasih **GEMINI API KEY** baru.

Input langsung ke terminal dengan mode hidden.

Cek secret AI yang dipakai

`kubectl describe deployment ufosi-ai -n ufosi | sed -n '/Environment:/,/Mounts:/p'`

Cari bagian seperti:

GEMINI_API_KEY from secret ufosi-ai-secret

nama secret-nya `ufosi-ai-secret`.

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl describe deployment ufosi-ai -n ufosi | sed -n '/Environment:/,/Mounts:/p'
Environment:
  FLASK_DEBUG:    0
  GEMINI_API_KEY: <set to the key 'GEMINI_API_KEY' in secret 'ufosi-ai-secret'> Optional: false
  PORT:          8000
Mounts:          <none>
```

```
itops1@ubuntu-itops1:~/ufosi-ops$ kubectl describe deployment ufosi-ai -n ufosi | sed -n '/Environment:/,/Mounts:/p'
```

Environment:

FLASK_DEBUG: 0

GEMINI_API_KEY: <set to the key 'GEMINI_API_KEY' in secret 'ufosi-ai-secret'>
Optional: false

PORT: 8000

Mounts: <none>

```
itops1@ubuntu-itops1:~/ufosi-ops$
```

Input key baru secara hidden

```
read -s -p "NEW_GEMINI_API_KEY: " NEW_GEMINI_API_KEY; echo
```

Paste key baru dari Developer. Terminal tidak akan menampilkan apa-apa. Itu normal.

Cek panjang key:

```
echo "NEW_GEMINI_API_KEY_LENGTH=${#NEW_GEMINI_API_KEY}"
```

Panjangnya masuk akal.

```
itops1@ubuntu-itops1:~/ufosi-ops$ echo "NEW_GEMINI_API_KEY_LENGTH=${#NEW_GEMINI_API_KEY}"  
NEW_GEMINI_API_KEY_LENGTH=53
```

Update Kubernetes Secret

```
kubectl create secret generic ufosi-ai-secret -n ufosi \  
--from-literal=GEMINI_API_KEY="$NEW_GEMINI_API_KEY" \  
--dry-run=client -o yaml | kubectl apply -f -
```

Simpan evidence metadata secret,:

```
kubectl get secret ufosi-ai-secret -n ufosi \  
> ~/ufosi-ops/evidence/day11-gemini-key-rotation/after-ai-secret-metadata.txt
```

Restart AI deployment

```
kubectl rollout restart deployment/ufosi-ai -n ufosi
```

Tunggu rollout:

```
kubectl rollout status deployment/ufosi-ai -n ufosi --timeout=240s \  
| tee ~/ufosi-ops/evidence/day11-gemini-key-rotation/ai-rollout-after-gemini-key-rotation.txt
```

deployment "ufosi-ai" successfully rolled out

Cek pod AI

```
kubectl get pods -n ufosi -o wide | grep ai \  
> ~/ufosi-ops/evidence/day11-gemini-key-rotation/after-ai-pod.txt
```

```
cat ~/ufosi-ops/evidence/day11-gemini-key-rotation/after-ai-pod.txt
```

ufosi-ai-xxxxx 1/1 Running

Validasi AI healthcheck setelah key diganti

```
kubectl delete pod curl-test -n ufosi --ignore-not-found
```

```
kubectl run curl-test -n ufosi \
  --image=curlimages/curl:latest \
  --restart=Never \
  -- sleep 3600
```

```
kubectl wait --for=condition=Ready pod/curl-test -n ufosi --timeout=90s
```

```
kubectl exec -n ufosi curl-test -- curl -i http://ufosi-ai:8000/api/v1/health \
> ~/ufosi-ops/evidence/day11-gemini-key-rotation/after-ai-health.txt
```

```
kubectl delete pod curl-test -n ufosi --ignore-not-found
```

Cek:

```
cat ~/ufosi-ops/evidence/day11-gemini-key-rotation/after-ai-health.txt
```

HTTP/1.1 200 OK

Validasi Service Lain Setelah Gemini Key Update

Frontend:

```
curl -I http://192.168.23.72:30080 \
>
~/ufosi-ops/evidence/day11-gemini-key-rotation/frontend-health-after-gemini-key-rotation.txt
2>&1
```

Backend:

```
curl -i http://192.168.23.72:30081/api/v1/health \
```

```
>
~/ufosi-ops/evidence/day11-gemini-key-rotation/backend-health-after-gemini-key-rotation.txt
2>&1
```

Database:

```
kubectl exec deployment/ufosi-db -n ufosi -- \
psql -U ufo_admin -d "ufo-db" -c '\conninfo' \
> ~/ufosi-ops/evidence/day11-gemini-key-rotation/db-conninfo-after-gemini-key-rotation.txt
```

Cek ringkas:

```
grep "HTTP/1.1"
~/ufosi-ops/evidence/day11-gemini-key-rotation/frontend-health-after-gemini-key-rotation.txt
grep "HTTP/1.1"
~/ufosi-ops/evidence/day11-gemini-key-rotation/backend-health-after-gemini-key-rotation.txt
cat ~/ufosi-ops/evidence/day11-gemini-key-rotation/db-conninfo-after-gemini-key-rotation.txt
```

Frontend: HTTP/1.1 200 OK
Backend: HTTP/1.1 200 OK
DB: connected to database "ufo-db"

Cek Known Issue AI 500 Setelah Gemini Key Baru

Sebelumnya backend log sempat menunjukkan:

AI service error 500
Internal server error during anomaly detection
Internal server error during financial analysis

Setelah Gemini key baru, ambil log baru.

```
kubectl logs deployment/ufosi-backend -n ufosi --tail=200 \
>
~/ufosi-ops/final-documentation/evidence-pack/doc-017-known-issue-review/backend-log-aft
er-gemini-key-update.txt 2>&1
```

```
kubectl logs deployment/ufosi-ai -n ufosi --tail=200 \
```

>

```
~/ufosi-ops/final-documentation/evidence-pack/doc-017-known-issue-review/ai-log-after-gemini-key-update.txt 2>&1
```

Cek apakah masih ada error AI:

```
grep -iE "error|invalid|failed|gemini|500" \
~/ufosi-ops/final-documentation/evidence-pack/doc-017-known-issue-review/backend-log-after-gemini-key-update.txt
```

```
grep -iE "error|invalid|failed|gemini|500" \
~/ufosi-ops/final-documentation/evidence-pack/doc-017-known-issue-review/ai-log-after-gemini-key-update.txt
```

Tidak keluar apa-apa, itu bagus.

Buat Summary Gemini Key Rotation

Jalankan:

```
cat > ~/ufosi-ops/evidence/day11-gemini-key-rotation/gemini-key-rotation-summary.txt
<<'EOF'
```

```
# UFOSI Gemini API Key Rotation Summary
```

```
## Scope
```

Update Gemini API key used by UFOSI AI/OCR service.

```
## Action
```

Kubernetes Secret for AI/OCR was updated with the new GEMINI_API_KEY.

AI/OCR deployment was restarted to load the new secret.

```
## Security Control
```

The raw Gemini API key was not printed in public documentation, not stored in screenshot evidence, and not included in Word/PPT files.

```
## Validation
```

- AI/OCR deployment rollout completed successfully.
- AI/OCR pod returned to Running state.
- AI/OCR internal healthcheck returned HTTP 200 OK.
- Backend healthcheck remained HTTP 200 OK.
- Frontend remained reachable.
- Database connection remained successful.

Result

Gemini API key rotation completed successfully for Demo/UAT.
EOF

Cek:

cat ~/ufosi-ops/evidence/day11-gemini-key-rotation/gemini-key-rotation-summary.txt

```
itops1@ubuntu-itops1:~/ufosi-ops$ cat ~/ufosi-ops/evidence/day11-gemini-key-rotation/gemini-key-rotation-summary.txt
# UFOSI Gemini API Key Rotation Summary

## Scope
Update Gemini API key used by UFOSI AI/OCR service.

## Action
Kubernetes Secret for AI/OCR was updated with the new GEMINI_API_KEY.
AI/OCR deployment was restarted to load the new secret.

## Security Control
The raw Gemini API key was not printed in evidence, not stored in public documentation, and not included in screenshots.

## Validation
- AI/OCR deployment rollout completed successfully.
- AI/OCR pod returned to Running state.
- AI/OCR service log shows healthcheck HTTP 200.
- AI/OCR internal healthcheck returned HTTP 200 OK.
- Backend healthcheck remained HTTP 200 OK.
- Frontend remained reachable.
- Database connection remained successful.

## Result
Gemini API key rotation completed successfully for Demo/UAT.
```

Terraform Evidence

7.2 Command yang dijalankan di VM 2

bukan di VM 1.

mkdir -p ~/ufosi-terraform-evidence-final

cd ~/ufosi-terraform-topology

Simpan struktur file:

```
find . -maxdepth 3 -type f | sort \
> ~/ufosi-terraform-evidence-final/terraform-file-structure.txt
```

Terraform version:

```
terraform version \
> ~/ufosi-terraform-evidence-final/terraform-version.txt
```

```
itops2@ubuntu-itops2:~/ufosi-terraform-topology$ terraform version
Terraform v1.15.5
on linux_amd64
+ provider registry.terraform.io/hashicorp/local v2.9.0
+ provider registry.terraform.io/hashicorp/null v3.3.0
```

Terraform init:

terraform init \

> ~/ufosi-terraform-evidence-final/terraform-init.txt 2>&1

```
itops2@ubuntu-itops2:~/ufosi-terraform-topology$ terraform init
Initializing provider plugins found in the configuration...
- Reusing previous version of hashicorp/local from the dependency lock file
- Reusing previous version of hashicorp/null from the dependency lock file
- Using previously-installed hashicorp/null v3.3.0
- Using previously-installed hashicorp/local v2.9.0

Initializing the backend...

Initializing provider plugins found in the state...
- Reusing previous version of hashicorp/local
- Reusing previous version of hashicorp/null
- Using previously-installed hashicorp/local v2.9.0
- Using previously-installed hashicorp/null v3.3.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
itops2@ubuntu-itops2:~/ufosi-terraform-topology$
```

Terraform validate:

terraform validate \

> ~/ufosi-terraform-evidence-final/terraform-validate.txt 2>&1

```
itops2@ubuntu-itops2:~/ufosi-terraform-topology$ terraform validate
Success! The configuration is valid.
```

Terraform plan:

terraform plan \

> ~/ufosi-terraform-evidence-final/terraform-plan.txt 2>&1

```

itops2@ubuntu-itops2:~/ufosi-terraform-topology$ terraform plan
null_resource.generate_evidence: Refreshing state... [id=4318614725301976576]
local_file.operational_summary: Refreshing state... [id=172180b49f7d1dd39d41df205e85d1bda2210f67]
local_file.firewall_matrix: Refreshing state... [id=1088937fb23163a2cdfc409950798e4e0dd513fb]
local_file.port_matrix: Refreshing state... [id=aaf1fee20cad0d4262653a9fe0308f3b9c8ada5b]
local_file.real_topology: Refreshing state... [id=576e77b69192050becf3e19bb20b31511f45f67a]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are
needed.

```

Tidak mau apply ulang, pakai existing log.

cat evidence/apply.log \

> ~/ufosi-terraform-evidence-final/terraform-apply-log.txt 2>&1

```

itops2@ubuntu-itops2:~/ufosi-terraform-topology$ cat evidence/apply.log
Terraform Applied: Tue Jun  9 07:29:42 AM UTC 2026
Project: UFOSI
Environment: Demo/UAT
VM1: ubuntu-itops1 192.168.23.72
VM2: ubuntu-itops2 192.168.23.71
Generated files:
total 16K
drwxrwxr-x 2 itops2 itops2 4.0K Jun  9 07:29 .
drwxrwxr-x 8 itops2 itops2 4.0K Jun  9 07:29 ..
-rwxrwxr-x 1 itops2 itops2 1.1K Jun  9 07:17 03-ufosi-firewall-matrix.md
-rwxrwxr-x 1 itops2 itops2 1.1K Jun  9 07:29 04-ufosi-operational-summary.md

```

Terraform output:

terraform output \

> ~/ufosi-terraform-evidence-final/terraform-output.txt 2>&1

```

itops2@ubuntu-itops2:~/ufosi-terraform-topology$ terraform output
environment = "Demo/UAT"
frontend_url = "http://192.168.23.72:30080"
generated_files = [
  "./generated/01-ufosi-real-topology.md",
  "./generated/02-ufosi-port-matrix.md",
  "./generated/03-ufosi-firewall-matrix.md",
  "./generated/04-ufosi-operational-summary.md",
]
grafana_url = "http://192.168.23.72:30300"
internal_services = {
  "ai_service" = "8000 -> target 8000"
  "backend_port" = 8080
  "database_port" = 5432
}
lab_subnet = "192.168.23.0/24"
project_name = "UFOSI"
prometheus_url = "http://192.168.23.72:30090"
terraform_scope = "Terraform is used as Infrastructure as Code evidence to generate topology documentation, port matrix,
firewall matrix, and apply evidence. Terraform does not provision VMware, GNS3, Cisco, or Mikrotik devices in this UFOS
I environment."
vm1_application_server = {
  "hostname" = "ubuntu-itops1"
  "ip" = "192.168.23.72"
  "role" = "Application Server: Docker, K3s/Kubernetes, UFOSI app, Prometheus, Grafana"
}
vm2_automation_server = {
  "hostname" = "ubuntu-itops2"
  "ip" = "192.168.23.71"
  "role" = "Automation Server: Terraform, IaC topology documentation, evidence generation"
}

```

Terraform state:

terraform state list \

> ~/ufosi-terraform-evidence-final/terraform-state-list.txt 2>&1

```
itops2@ubuntu-itops2:~/ufosi-terraform-topology$ terraform state list
local_file.firewall_matrix
local_file.operational_summary
local_file.port_matrix
local_file.real_topology
null_resource.generate_evidence
```

Buat topology summary:

```
cat > ~/ufosi-terraform-evidence-final/terraform-topology-summary.txt <<'EOF'
# UFOSI Terraform Topology Evidence Summary
```

Scope

Terraform is used as network topology automation and governance evidence for UFOSI Demo/UAT.

Environment

Host: VM 2 / ubuntu-itops2

Role: Terraform & Governance Operator

Purpose

Terraform is used to document and simulate network topology automation, including LAN topology, connectivity validation script, and optional Mikrotik/GNS3 topology evidence.

Limitation

Terraform is not used as full production infrastructure provisioning in this project. It is used as topology automation evidence and governance support for Demo/UAT.

Expected Evidence

- terraform init
- terraform validate
- terraform plan
- terraform apply log
- terraform output
- terraform state list
- generated topology/readme files

EOF

Zip evidence di VM 2:

```
cd ~
```

```
zip -r ufosi-terraform-evidence-final.zip ufosi-terraform-evidence-final
```

```
ls -lh ~/ufosi-terraform-evidence-final.zip
```
